

SLA 3D Printing

Parameters on Surface Quality and Dimensional Accuracy

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MAE 259: Modern Manufacturing Technologies

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March 16th, 2025

Abstract

Stereolithography (SLA) printing utilizes UV light to cure resin in layers, creating solid objects. Parameters such as layer height and layer exposure time can significantly affect the total print time, as well as the surface finish and dimensional accuracy of the final printed part. By printing fourteen batches with five samples each while varying the layer height and layer exposure time we sought to optimize the SLA manufacturing process for high accuracy, low surface roughness and reduced printing time. Optimizing these parameters will allow for quicker, but still high quality, prints, increasing the total production rate. Measurements were taken with gauge pins and calipers, and surface finish was evaluated using a surface roughness tester, the Mahr MarSurf PS10.

Comparing the measurement error from the designed part and the surface finish to the rate of failed prints and total print time, allowed us to determine several conclusions. First, layer thickness had a direct linear relationship with surface roughness, while exposure time did not have any clear relationship, indicating that surface finish can be controlled by adjusting only the layer height. Second, layer exposure time had a logarithmic relationship to measurement accuracy, with longer exposure providing lower standard deviation from the design. Finally, since layer exposure time and layer height both affected the failure rate, we were able to determine that, if surface finish requirements allow, a 45 μ m layer height at 4 seconds exposure gave the shortest print time with the least failures and the lowest measurement deviation.

Introduction

Stereolithography (SLA) is one of the earliest forms of 3D printing, developed in the 1980s and patented in 1986 (Hull, 1986). It utilizes an ultraviolet (UV) laser to selectively cure liquid photopolymer resin, layer by layer, into solid objects. The technology has gained popularity due to its ability to produce highly detailed and smooth-surfaced prints, making it a preferred choice for applications requiring precision and intricate designs. However, various print parameters can significantly affect the final part's surface quality, dimensional accuracy, and print speed.

This study investigates the effects of changing SLA 3D print parameters, specifically layer thickness and exposure time, on the surface roughness and dimensional accuracy of printed parts. The research aims to determine optimal settings that balance print quality, efficiency, and reliability. Measurements were conducted using calipers, gauge pins, and a surface roughness tester to evaluate the impact of different print settings.

Figure 1 shows the SLA 3D printing process using a laser focused on a point in a photopolymer resin. The build plate moves vertically, and the part is printed

upside-down as it moves up by the determined layer height. Figure 2 shows the printing process using a Masked Stereolithography (MSLA) printer, which uses an LCD screen to dynamically mask areas from being exposed to UV light, allowing the entire layer to be cured at once. This technology increases print speed with the primary limitation of print speed being controlled by the total print height.

Figure 1: SLA 3D Laser Printing Diagram

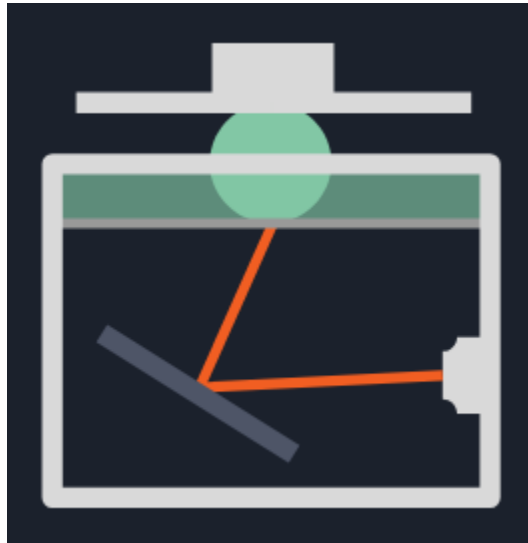
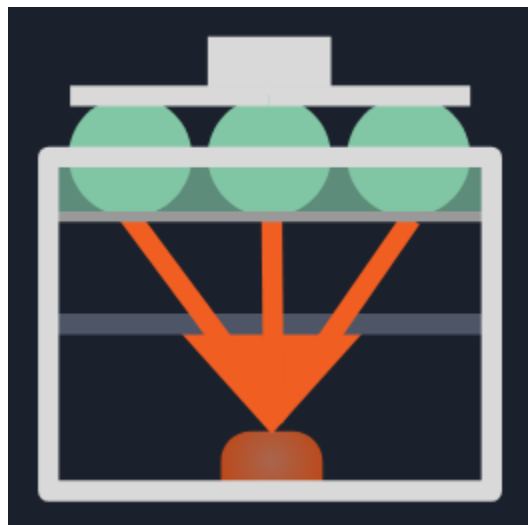


Figure 2: SLA LCD 3D Printing



SLA 3D Printing Market

SLA printing is used for prototyping in the healthcare, automotive, aerospace, dentistry, jewelry, and other industries for rapid prototyping. SLA allows high surface quality parts to be printed, including using biologically compatible materials. The SLA 3D printing market in 2021 was \$1.964 billion globally, and is projected to increase to \$6.746 billion in 2028 (Knowledge Sourcing Intelligence).

The versatility of SLA 3D printing has led to its widespread adoption across various industries. In healthcare, it allows the creation of patient-specific medical devices and anatomical models, enhancing surgical planning and personalized treatment. The automotive and aerospace sectors utilize SLA for rapid prototyping and the production of lightweight, durable components, streamlining product development cycles and reducing time-to-market. The consumer goods industry leverages SLA technology for designing and manufacturing customized products, catering to the growing demand for personalization. As technological advancements continue and material costs decline, the SLA 3D printing market is poised for sustained expansion, solidifying its role as a critical component in modern manufacturing processes (Zongheng 3D).

Origins and Fundamentals of SLA 3D Printing

SLA 3D printing was pioneered by Charles Hull, who later founded 3D Systems, a leading company in additive manufacturing. SLA is a vat photopolymerization process in which a UV laser selectively solidifies resin in a tank, building the part layer by layer. The process consists of several critical components:

1. **Resin Tank** – A container filled with liquid photopolymer resin.
2. **Laser and Galvanometers** – A UV laser beam directed by galvanometers to cure resin selectively.
3. **Build Platform** – The platform raises incrementally as each layer solidifies.
4. **Software and G-Code Generation** – Slicing software generates instructions for laser movement and exposure times.

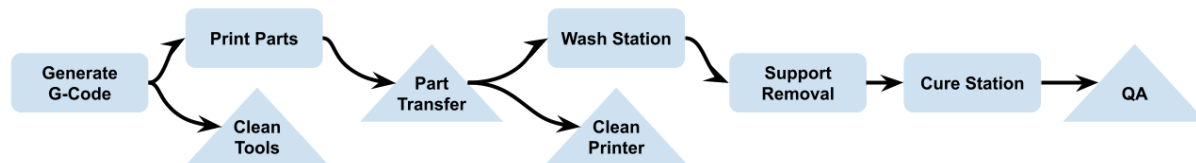
SLA offers advantages such as high accuracy, excellent surface finish, and suitability for post-processing (e.g., machining and polishing). However, it also has drawbacks, including high material costs, required post-curing steps, and sensitivity to environmental conditions.

The SLA Manufacturing Process

The SLA printing process involves several key steps, each influencing the final part's accuracy and quality:

1. **Generating Stereolithography (STL) file** - A 3D model is sliced into layers of a specified thickness, and then the layers, along with the print settings, are compiled into an STL file.
2. **Printing** - After generating the STL file, the print bed is filled with resin, and the STL is loaded into the printer. The print is then started, and the tools are wiped down with 99% isopropyl alcohol to remove any uncured resin.
3. **Wash Station** - Parts are removed from the build plate and transferred to a wash station, where they are submerged in 99% isopropyl alcohol. The alcohol is agitated using an impeller to remove any uncured resin from the printed parts. Any remaining resin in the print bed is poured back into the resin bottles, and all contaminated surfaces are sprayed with 99% isopropyl alcohol and wiped down with paper towels.
4. **Support Removal** - Parts are rinsed with water, and the supports are carefully removed from the printed parts.
5. **Cure Station** - Parts are placed on a rotating bed under ultraviolet light to post-cure the parts.
6. **Quality Assurance** - From there, parts are ready to be inspected for any defects or deviations from intended dimensions [Figure 3]

Figure 3. Manufacturing process of an SLA 3D printed part



Methods

For printing our samples we used the Anycubic Photon M3 MSLA printer, with Anycubic Standard beige resin. This printer uses a matrix of LEDs which shine 405nm UV light through a 4K+ 7.6" monochrome LCD screen. The Anycubic Wash & Cure Plus machine was used to clean parts and perform the final cure process, using a rotating cure plate and 405nm UV lamp beads to ensure an even cure. A 250ml glass beaker was used to measure out the resin added before each batch, adding 150ml before each batch was printed. A steel scraper was used to remove parts from the baseplate, and a plastic scraper was used to remove excess cured and uncured resin from the bed between prints. A steel funnel with a perforated steel sieve was used to return uncured resin back into the bottle during the cleaning process while separating out the cured resin. All tools were cleaned with a spray bottle filled with 99% isopropyl alcohol and paper towels.

Since printing time is the largest limit in the manufacturing process, two parameters were chosen that would significantly affect the total print time:

1. **Layer Thickness:** Adjusted in increments of 5 micrometers (μm) from 25 μm to 45 μm .
2. **Layer Exposure Time:** Adjusted in 1 second increments from 2 to 4 seconds per layer.

Fifteen total batches were printed, with five samples in each batch. The samples all used identical support structures, and were labeled to indicate the location on the bed each sample was printed on. Samples were rotated with a 15 degree rotation in the x-axis and the y-axis, both to reduce total peeling force (Huang, 2005) and to better allow resin to drain off of the samples and back into the print bed. Anti-aliasing was not enabled, and all other parameters were left at their default values for every batch [Figure 4].

Figure 4. Print settings for a batch with 25 μm layer height and 4 second exposure time.

The screenshot shows the 'Anycubic Rapid / Basic / Standard Grey 25 um' settings window. It includes fields for Brand (Anycubic), Name (Rapid / Basic / Standard), and Color (Grey). The 'Burn In Layers' section has settings for Number of Layers (2), Exposure Time (40 s), Transition Layers Count (6), Lift Distance (4 mm), Lift Speed (65 mm/m), and Retract Speed (150 mm/m). The 'Normal Layers' section has settings for Layer Thickness (25 um), Light-off Delay (1 s), Exposure Time (4 s), Lift Distance (4 mm), Lift Speed (65 mm/m), and Retract Speed (180 mm/m). The 'Scene Scale compensation' section has Scale X (100%), Scale Y (100%), and Scale Z (100%). The 'Compensation' section has a 'Plus' button and a 'Support resin compensation' toggle (on/off). The 'Price' section has fields for Bottle Price (25.99 USD) and Bottle Capacity (1 l). The 'Print Time Override' section has a toggle (on/off) and fields for Time per Burn In layer (0 s) and Time per layer (0 s). At the bottom, there are 'Delete Resin' and 'OK' buttons.

Inspection Methods

To assess the impact of these parameters, several measurement techniques were employed:

- **Calipers:** Used to measure length, width, hole diameter, and the distance between features.

- **Gauge Pins:** Determined the actual hole size by fitting the largest possible pin.
- **Surface Roughness Tester (Mahr MarSurf PS10):** Measured the arithmetic average roughness (Ra) using a 2- μ m probe tip.

The combination of these tools allowed for a comprehensive evaluation of how each parameter affected both geometric accuracy and surface quality.

Calipers were used to measure overall dimensions of the part, distance between the cylinders, and the distance between the holes as seen in the figure below. Mitutoyo calipers like those used have a measurement tolerance of ± 0.001 " (0.02 mm). The way that the calipers were applied, as seen in the figure, means that the highest points of either face are measured. The jaws of the calipers were carefully positioned to ensure they made firm contact with the reference points on the part without applying excessive force, which could lead to deformation or misreading. The figure below illustrates a measurement of the overall length, where the caliper jaws are aligned with the outermost edges of the part. This method ensures that the recorded dimension represents the maximum length, accounting for any irregularities or surface variations in the material. Calipers were chosen as an acceptable tool considering the dimensional tolerance of the printer.

Figure 5: Caliper measurement of overall length



The gauge pins used were a metric set in increments of .02 mm. These precision-ground cylindrical pins allow for accurate hole measurement by determining the maximum pin size that can fully insert without excessive force. When measuring the hole size, the smallest pin was first inserted, and if it fit easily, progressively larger pins were tested until the largest one that could fully enter the hole was determined. This final pin size was recorded as the hole's actual diameter. This process was repeated for both left and right holes. The gauge pin should not be inserted with excessive force, to ensure the insertion does not remove the softer plastic material around the hole. The gauge pins eliminate subjective readings or circularity errors because they provide a direct physical fit and measurement.

Figure 6: Gauge pins used for hole measurement of left hole



The Mahr MarSurf PS10 was used for surface measurement of one part of each batch. As shown below, the 2 μm stylus probe tip was placed roughly in the center of the part, perpendicular to the long edge. When the measurement button on the tool is pressed, the probe traverses a total length of 15mm along the part's surface and then returns to its starting position. During this process, the stylus tip moves up and down, recording the variations in surface height. The measuring force applied by the probe is 0.00075 N, ensuring minimal interference with the surface texture while capturing accurate data. The probe moves at a controlled speed of 1 mm/s, which allows for precise surface characterization (Mahr Metrology).

The tool reports R_a , average surface roughness. R_a is the average height of the peaks and valleys of the surface across the measured section (Stahli USA). This value provides an indication of the texture and smoothness of the part, which is critical for ensuring proper functionality in applications requiring precise surface finishes. As illustrated in Figure 8, the stylus probe tip size directly influences the measurement's accuracy and resolution. The probe tip acts as a mechanical filter, meaning that it can only detect features larger than its own diameter. If the actual surface profile contains finer features or sharper peaks and valleys smaller than 2 μm , they may not be accurately recorded. This phenomenon, known as aliasing, can result in an incomplete or misleading surface roughness measurement. When using the surface measurement tool, care must be taken not to allow the probe to fall off the surface where it can be damaged.

Figure 7: Mahr MarSurf PS10 Surface Measurement Tool

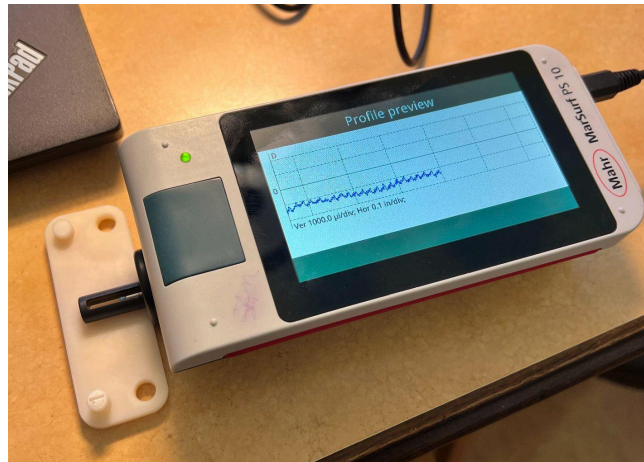
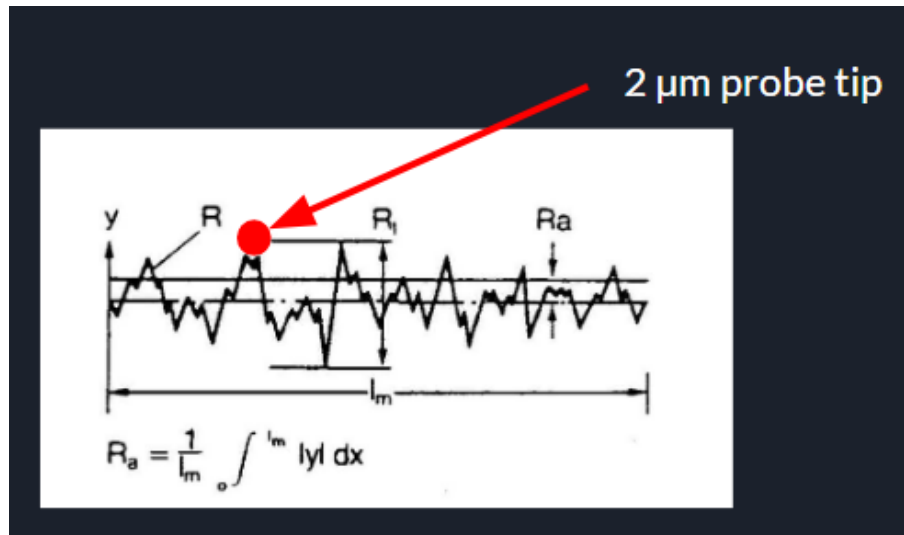


Figure 8: Surface roughness with simulated probe tip



The Photon M3 displays the final print time when it finishes printing, so that was used to record the total time taken to print each batch and ambient room temperature was taken at the end of each print using the readout on a fan being used for ventilation.

Results and Findings

The study found several key correlations between print parameters and part quality:

1. Surface Roughness:

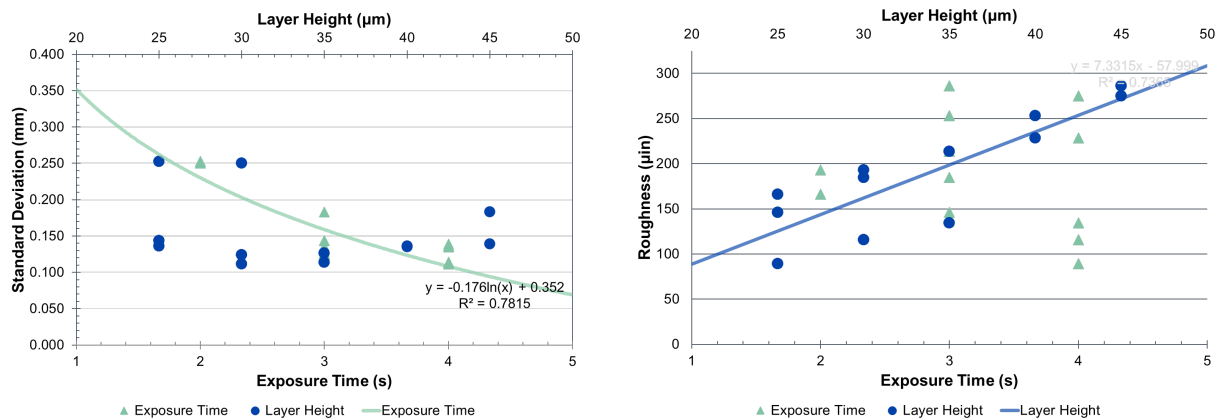
- A positive correlation was observed between layer height and roughness. Thicker layers are directly related to higher R_a values, indicating rougher surfaces.

- Exposure time did not have a clear correlation with surface roughness, exhibiting significant scatter. [Figure 9]

2. Dimensional Accuracy:

- Measurement deviation increased as exposure time decreased. Shorter exposure times led to under-cured layers, resulting in inconsistent dimensions.
- Layer height appears to have little impact on measurement accuracy. While lower layer height did show an increase in scatter, more samples would be needed to determine a clear relationship [Figure 9]

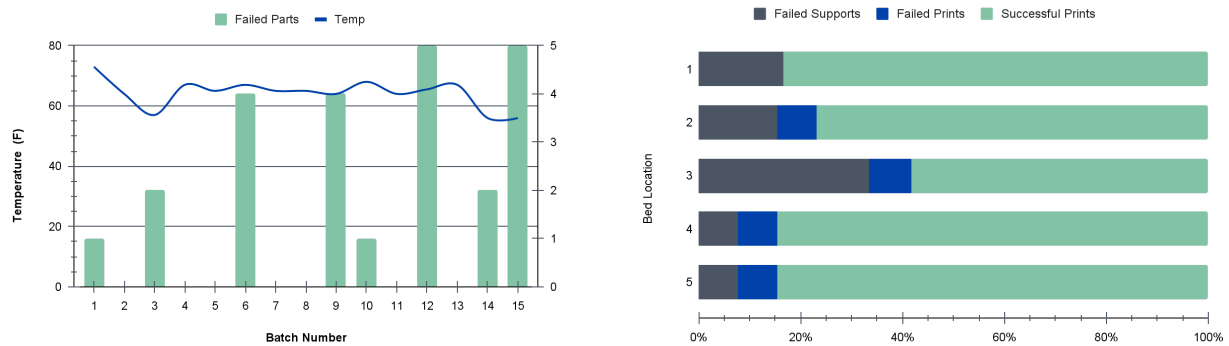
Figure 9. Charts showing the relationship between measurement deviation versus layer height and exposure time (left) and the relationship between surface roughness versus layer height and exposure time (right)



Failure Analysis and Recommendations

Since temperature can affect the viscosity of resin, the ambient room temperature at the end of each print was compared to the number of failed prints in each batch [Figure 10]. We did see a slight increase in failures at the two lowest recorded temperatures, however, more samples with a higher temperature sample rate are needed to determine a strong relationship between ambient temperature and failure rate [Figure 10].

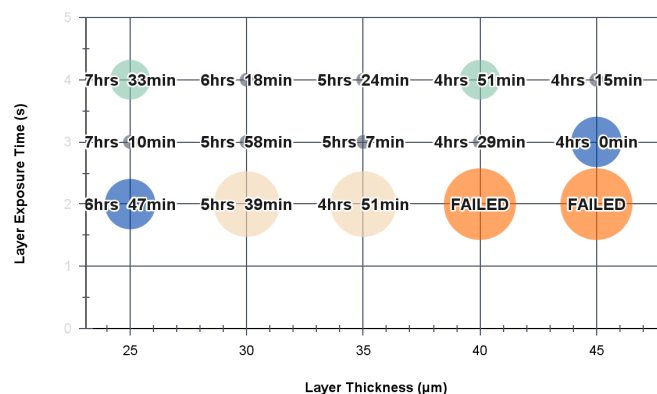
Figure 10. Chart showing the number of failed samples in each batch combined with the ambient air temperature (left), and chart showing failure rate as a percentage of samples in each bed location (right)



Print failures were also compared to the location of each sample on the print bed [Figure 10]. The center of the bed did show slightly more failed samples than the other four locations, which could indicate that avoiding the center of the print bed will decrease the chance of failure. However, due to a relatively small sample size of 14 batches, more tests are needed to increase confidence in this result. Further testing on multiple machines would also be needed to determine if this is a flaw in this specific printer or if it is a result of the design of the Anycubic Photon M3 print bed.

Layer height showed a direct relation to failure rate [Figure 11], though it was only visible at two and three seconds layer exposure, with the larger layer heights showing slightly increased number of failed samples. Exposure time, however, showed a strong correlation with failed prints, with all batches at a two second exposure time having samples that failed to print [Figure 11]. More samples at smaller exposure time intervals are needed to learn more about the exact relationship between layer exposure time and print failures.

Figure 11. Chart showing the number of failed samples at each layer height and exposure time, with bigger bubbles indicating more failed prints



From these findings, several recommendations for SLA manufacturing were determined:

- **Surface Finish Optimization:** Layer height can be adjusted to control the final surface roughness to achieve a desired surface finish, with smaller layers decreasing surface roughness.
- **Dimensional Accuracy Improvement:** To minimize measurement deviation, a longer exposure time (4 seconds) is recommended.
- **Time-Efficient Printing:** A 45µm layer height with 4 second exposure time provides the best balance of speed and reliability, taking only 6% longer and having zero failed prints compared to a 3 second layer exposure at the same layer height, which had a 60% failure rate.

Future Studies and Potential Improvements

Several areas for future research have been identified:

1. **Fine-Tuning Exposure Times:** Investigating exposure times between 2 and 3 seconds to determine the precise threshold for failure. By increasing the number of steps you can further optimize print time and quality.
2. **Anti-Aliasing Mode Analysis:** Evaluating how software-based anti-aliasing affects roughness and accuracy. Since anti-aliasing is meant to soften hard edges on curves, it may help increase the surface finish, though that may have an adverse effect on the measurement deviation.
3. **Batch Analysis:** Printing multiple samples with identical settings to quantify repeatability and establish statistical confidence in the findings. With a higher number of samples it will be easier to identify any outliers.
4. **Viscosity Analysis:** Test SLA printing with different ambient temperatures to the failure rate due to resin viscosity. By controlling the ambient temperature, more accurate data could be found for the relationship between failure rate and ambient temperature.

Conclusion

Because of its increased prominence in industry, understanding SLA 3D printing parameters effects on part features, print time, and failure rates is increasingly important. To ensure that rapid prototyping is cost effective, print speed should be optimized to achieve the desired outcome. The findings of this report are important for understanding how surface quality and dimensional accuracy is affected by the print parameters that also most significantly impact print time. Future studies can increase this body of knowledge, allowing SLA 3D printing operators to minimize print time and increase throughput while still meeting needed physical attributes.

This study demonstrated that SLA print quality and accuracy depend significantly on layer thickness and exposure time. While thinner layers improve surface quality, they also increase print time and can lead to inconsistencies in small features. Longer exposure times enhance dimensional accuracy at the cost of increasing print time. By balancing the savings from a shorter print time with the need for surface quality and measurement accuracy, one can determine the ideal printing temperature for their process. The findings provide valuable insights into optimizing SLA printing parameters for specific applications, helping manufacturers balance quality, efficiency, and material costs. Continued exploration of additional variables, such as anti-aliasing and resin viscosity, could further refine the SLA printing process.

References

- Hull, C. (1986). *Apparatus for production of three-dimensional objects by stereolithography* (U.S. Patent No. 4575330). U.S. Patent and Trademark Office.
<https://patents.google.com/patent/US4575330A/en>
- Huang, Y., & Jiang, C. (2005). On-line force monitoring of platform ascending rapid prototyping system. *Journal of Materials Processing Technology*, 159(2), 257-264.
<https://doi.org/10.1016/j.jmatprotec.2004.05.015>
- Knowledge Sourcing Intelligence. (n.d.). Stereolithography (SLA) technology 3D printing market report. *Knowledge Sourcing Intelligence*
<https://www.knowledge-sourcing.com/report/sterolithography-sla-technology-3d-printing-market>
- Mahr Metrology. (n.d.). Mobile roughness measuring device – MarSurf PS 10. *Mahr Metrology*.
<https://metrology.mahr.com/en-int/products/article/6910230-mobiles-rauheitsmessgeraet-marsurf-ps-10/>
- Stahli USA. (n.d.). Surface finish quality – The technique of lapping. *Stahli USA*.
<https://stahlusa.com/stahli-publication/the-technique-of-lapping/surface-finish-quality/>
- Zongheng 3D. (n.d.). In what industries is SLA 3D printing commonly used? *Zongheng 3D*.
<https://www.zongheng3d.com/in-what-industries-is-sla-3d-printing-commonly-used/>